AI-Enabled Food Demand Forecasting and Waste Reduction System for Sri Lankan Restaurants

***Abstract*— Small-to medium-sized restaurants in Sri Lanka face uncertainty about demand due to weather and cultural events. They rely on manual judgment for inventory planning, causing stockouts and waste. This study developed an AI-enabled system for demand forecasting and inventory control, predicting menu demand, converting forecasts to ingredient procurement, and improving buffer policies using waste data. The architecture includes a hybrid forecasting module using LightGBM and Prophet models with temporal features, refined by a weighted fuzzy-logic contextual engine using OpenWeatherMap data and a Sri Lankan cultural calendar. Forecasts are converted into ingredient needs via recipe mapping with an adaptive buffer algorithm that updates the safety stock using waste data and exponential smoothing. Implemented with a Next.js/React frontend and Django backend, the system was evaluated using MAE/RMSE/MAPE over 30 days. The results show improved forecast accuracy and reduced food waste compared with static-buffer methods, while maintaining low stockouts. This suggests that context-aware forecasting and adaptive control offer sustainable and efficient operations in small business settings, despite data quality and deterministic cultural rules.**

Keywords: Demand Forecasting, Hybrid Machine Learning, Food Waste Reduction, Adaptive Inventory Control Demand Forecasting, Hybrid Machine Learning, Food Waste Reduction, Adaptive Inventory Control

I. INTRODUCTION

The food service sector faces inherent challenges with unpredictable demand, a problem that is especially evident in small-to medium-sized restaurants in Sri Lanka. In these local settings, the number of customers and demand for specific menu items vary significantly owing to dynamic influences such as weather changes and complex schedules of cultural events and public holidays. Currently, inventory management in this industry largely depends on manual judgment and past records. The absence of data-driven strategies frequently results in two operational extremes: shortages that harm customer satisfaction and overproduction that leads to significant food wastage.[1]

Although digital systems for inventory management are available, they often do not align with the practical needs of small-scale businesses. Many traditional solutions offer isolated forecasting, attempting to predict future sales without converting these forecasts into practical, ingredient-specific production plans. Additionally, these systems function independently and lack essential operational feedback mechanisms. They overlook the actual daily food waste produced in the kitchen, resulting in inventory buffer policies that remain unchanged and fail to adjust to real world inefficiencies over time.[2]

This study seeks to overcome these challenges by creating a comprehensive AI-enhanced system for demand forecasting and inventory management tailored to the unique needs of restaurants in Sri Lanka. The proposed framework surpasses conventional forecasting methods by operating as a complete pipeline that employs a streamlined hybrid machine learning model to forecast menu demand using temporal, cultural, and weather-related data. These forecasts are then translated into accurate ingredient requirements. Importantly, this system introduces an innovative adaptive buffer algorithm at the ingredient level. By tracking actual unsold food waste from the kitchen, a waste-driven feedback mechanism was established that automatically fine-tunes safety stock levels, effectively linking predictive analytics to everyday kitchen operations.

This study is significant because it introduces a sustainable approach to restaurant management. By transitioning from manual estimates to a flexible, data-driven framework, the proposed system alleviates the financial burden of overpurchasing, enhances overall operational efficiency, and provides a scalable and technologically accessible solution for reducing localized food waste in the food service industry.

The remainder of this paper is organized as follows. Section II examines related studies and the current literature on forecasting restaurant demands. Section III outlines the system architecture and complete data flow process. Section IV describes the methodology, which includes the hybrid forecasting model and adaptive buffer algorithm. Section V presents the experimental findings and evaluates the performance. Finally, Section VI concludes the paper and explores potential directions for future research.

II. RELATED WORK

A. AI and Machine Learning in Food Demand Forecasting

Extensive research has been conducted on precise demand forecasting in the food service industry, evolving from traditional statistical techniques to sophisticated machine learning (ML) models. Initially, methods such as ARIMA and exponential smoothing were heavily utilized for time-series forecasting; however, they often fell short of capturing the nonlinear and unpredictable nature of restaurant sales [3]. Recent studies have emphasized the effectiveness of ML algorithms, especially tree-based models such as Random Forest and LightGBM, along with specialized time-series forecasting tools such as Facebook Prophet, in managing complex multivariate data [4],[5]. Nonetheless, a major drawback of current forecasting research is its dependence on large datasets, which renders these models impractical for

smaller firms to use. Additionally, while some studies include general calendar features, few have focused on incorporating highly localized, intricate cultural events, such as specific Sri Lankan public holidays and festivals, into the main predictive model [6].

B. Inventory and Reorder Management in the Food Service Sector

Traditionally, research on inventory management in the food sector has focused on refining the Economic Order Quantity (EOQ) and maintaining static safety stock levels [7]. With the advent of modern digital inventory systems, these concepts have been digitized, enabling automated notifications for reordering when the inventory dips below a set threshold [8]. Although these systems enhance administrative efficiency, they rely on inflexible and static policies. Studies have indicated that static safety stocks are inadequate for addressing the perishable nature of restaurant ingredients and the variable daily demands of kitchen operations [9].

C. Identified Gaps and Research Positioning

A critical review of the existing literature reveals two primary research gaps. First, there is a distinct lack of integration between predictive demand forecasting and inventory management at the ingredient level. Most existing systems treat forecasting as a standalone module, failing to dynamically convert predicted menu sales into actionable recipe-mapped ingredient requirements [10]. Second, the current literature lacks mechanisms for operational feedback; inventory buffers are rarely adjusted based on actual kitchen waste in real time. This study addresses these gaps by introducing a comprehensive, end-to-end framework. Unlike isolated forecasting tools, the proposed system utilizes a streamlined hybrid machine-learning and rule-based design that seamlessly converts context-sensitive forecasts into procurement strategies that can be implemented. Additionally, it presents an innovative adaptive inventory buffer algorithm that persistently adjusts target stock levels based on actual waste feedback, creating a dynamic connection between the anticipated demand and real-time operational performance.

III. SYSTEM ARCHITECTURE

The proposed system architecture is structured as a comprehensive pipeline featuring two main components: a predictive forecasting engine and an operational inventory management system. Unlike isolated predictive models, this architecture establishes a continuous feedback loop in which the anticipated demand informs production planning and actual operational waste adjusts future inventory reserves.

A. High-Level System Data Flow

The operational pipeline begins with the storage of daily sales transaction data. These records, along with temporal and meteorological features, were fed into the forecasting module to predict future menu requirements for the canteen. The predicted demand is then parsed by the kitchen production planning module, which maps menu item quantities to their specific ingredient requirements using defined recipes. The inventory module calculates the required purchases based on these ingredient projections and dynamic buffer levels, passing actionable orders to suppliers. Finally, real unsold waste is captured and fed back into the system to automatically update the safety-buffer policy.

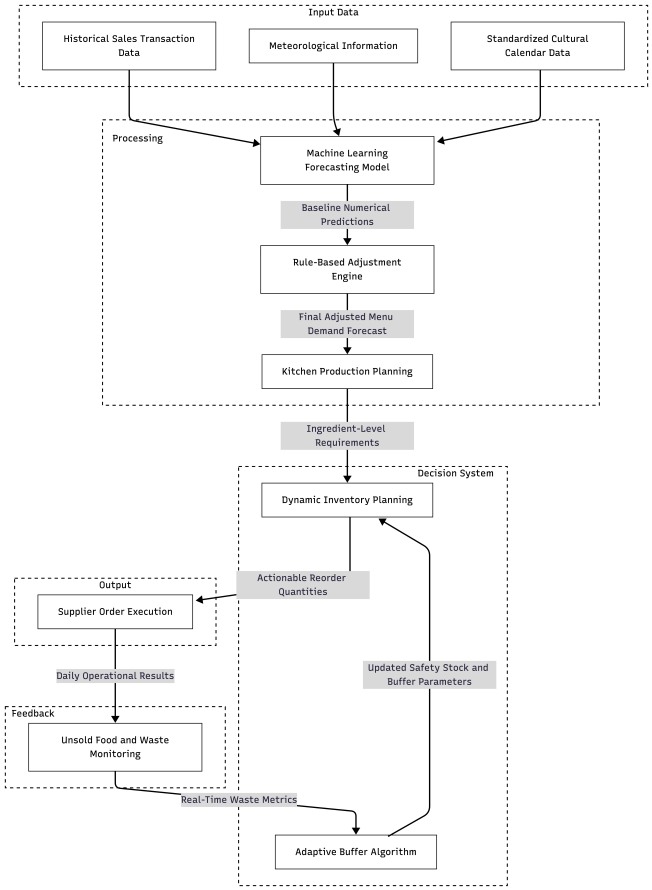

Fig. 1. Architecture of the forecasting and inventory management system.

B. Lightweight Hybrid Food Demand Forecaster

The first core component is a hybrid forecasting module tailored for small datasets typical of localized restaurants. It uses a two-tier approach to generate the predictions.

- ML Base Prediction: A lightweight machine learning model—utilizing LightGBM, XGBoost, or Facebook Prophet—is trained on historical sales, day-of-the-week, month, and lag features to generate a baseline numerical forecast.
- Rule-Based Adjustment Engine: To account for high-variance cultural and environmental events, the baseline forecast is passed through a deterministic if-then-else rule engine. This engine ingests automated meteorological data (e.g., precipitation and temperature via OpenWeatherMap) and a hardcoded Sri Lankan cultural calendar. The final adjusted demand forecast is calculated as follows:

$$\hat{Y}_{final} = \hat{Y}_{base} \times \frac{\sum_{i=1}^{Z} w_i \delta_i}{\sum w_i}$$

In this equation, $\hat{Y}_{\text{final}}$ represents the calibrated baseline prediction generated by the ML core, which incorporates the lagged and rolling features. The term δ_i denotes the fuzzy adjustment factor corresponding to each contextual event, while w_i represents the assigned weight that reflects the historical reliability of the respective rule. This formulation ensures a mathematically grounded forecasting approach tailored to the Sri Lankan environment.

C. AI-Supported Inventory and Reorder Management

The second core component translates the optimized forecast into physical inventory actions. The system is built on a robust, scalable stack: a Next.js and React frontend utilizing TypeScript for the user interface, paired with a Django REST Framework (DRF) backend. The underlying data were managed in a PostgreSQL database, whereas Python-based ML models were serialized and loaded via joblib into the backend service.

- Dynamic Inventory Planning: The system computes the total required ingredient quantity from the menu forecasts. The target stock level is calculated dynamically as follows:

$$S_{target} = L_{reorder} + B_{size}$$

  Where S_{target} is the target stock level, $L_{reorder}$ is the baseline reorder level, and B_{size} is the active buffer size. The system then evaluates the projected remaining stock to categorize the risk status and automatically suggests the purchase quantities.
- Adaptive Buffer Algorithm: To prevent static overpurchasing, the system features a waste-driven feedback mechanism. Menu waste is converted into ingredient-equivalent waste by recipe mapping. The algorithm calculates the target buffer based on the average daily waste (W_{avg}) over a configurable lookback window and the desired buffer days (D_{buffer}) as follows:

$$B_{target} = W_{avg} \times D_{buffer}$$

  To ensure operational stability and prevent volatile swings in purchasing, the system applies an exponential smoothing mechanism to update the final buffer policy as follows:

$$B_{new} = B_{old}(1-\alpha) + B_{target}\alpha$$

  where α is the smoothing factor that governs the update rate. This allows the system to automatically adjust the safety buffers based on real operational inefficiencies rather than rigid assumptions.

IV. IMPLEMENTATION

This section describes the practical implementation of the system, including the processes involved in data preparation, technical setup used for deployment, and metrics designed to assess the predictive accuracy and operational viability of the system.

A. Dataset Collection and Preprocessing

To develop the hybrid forecasting module and evaluate the adaptive inventory buffer, we assembled a dataset from the historical records of small-to medium-sized Sri Lankan food service businesses. This primary dataset includes daily sales data, detailed menu transactions, and supplier invoices for a continuous 24-month period. Contextual information was incorporated by embedding a standardized Sri Lankan cultural calendar to monitor the public holidays. Daily weather data (temperature and precipitation) were obtained from the OpenWeatherMap API. Extensive preprocessing is required to align the data for time-series forecasting. The raw sales data uploaded manually via CSV files were cleaned to rectify missing values and errors caused by manual logbook entry. The dates were standardized, and the sales per item were aggregated daily. Feature engineering was subsequently employed to derive essential temporal indicators (such as day_of_week, month, and is_weekend) and to calculate rolling statistical features (such as lag 1 and lag 7) to capture sequential dependencies before model training.

B. Hardware and Software Environment

The system was created and assessed using a strongly decoupled software architecture built for scalability and minimization of operational expenses.

- Software Stack: The backend system was constructed using the Django REST Framework (DRF) and connected to a PostgreSQL database. The machine learning component, which handles the LightGBM and Facebook Prophet models, was created in a Python environment using the Pandas and Scikit-learn libraries. The final models were serialized and integrated into the backend using the Joblib library. The front-end interface was developed using Next.js, React, and the TypeScript programming language.
- Hardware Setup: The training of models and backend testing were performed on a machine featuring an AMD Ryzen 7 processor, 16GB RAM, and an NVIDIA RTX 3060.

C. Evaluation Metrics

A dual-metric evaluation strategy was used to ensure that the system achieved both predictive and operational goals.

- Predictive Accuracy: The effectiveness of the LightGBM and Prophet baseline models was assessed using the Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). These metrics measure the average size of forecasting errors, allowing for straightforward comparisons with traditional moving average baseline models.

- Operational Performance: The practical efficiency of the web application and inventory module was gauged using two metrics. System responsiveness was assessed using API latency to ensure that the Next.js dashboard could access complex inventory risk statuses (OK, Low, and Stock Out) in real time. Furthermore, the success of the adaptive buffer algorithm was determined by calculating the percentage reduction in daily physical food waste compared to the frequency of inventory stockouts in a 30-day simulation.

V. RESULTS AND DISCUSSION

This section examines the performance of the proposed AI-assisted system in forecasting demand and managing inventory. The analysis considers both the predictive precision of the hybrid model and the operational effects of the adaptive inventory buffer on multiple restaurant datasets.

A. Evaluation Setup and Datasets

To assess the proposed system's efficiency, data on past sales, inventory, and waste were gathered from a dataset comprising 1500 small-to medium-sized restaurants in Sri Lanka over 24 or 48 months. The dataset was divided into a training set, which accounted for 80% of the data, and a testing set, which comprised the remaining 20% of the data. A simulated 30-day operational period was employed to examine the reaction of the dynamic inventory buffer to actual kitchen waste.

B. Forecasting Accuracy Results

The accuracy of the predictions made by the proposed hybrid model, which combines a Machine Learning baseline with a Rule-Based Adjustment Engine, was evaluated against standalone ML models, such as Light Gradient Boosting Machine and Prophet, which lack rule-based adjustments, as well as traditional moving average techniques. The evaluation metrics used were the Mean Absolute Percentage Error (MAPE) and Root Mean Squared Error (RMSE).

As shown in Table **??**, the hybrid model outperformed the standalone models. By integrating the Sri Lankan cultural calendar and weather data using a rule-based engine, the hybrid model effectively detected demand surges during specific events, such as Vesak and Sinhala New Year, which standalone machine learning models did not predict accurately.

TABLE I
PERFORMANCE HYBRID MODEL

Model Architecture	MAE	RMSE	R^2	MAPE (%)
Proposed Hybrid Model	3.42	4.89	0.88	12.5

TABLE II
PERFORMANCE COMPARISON OF TWO BASE MODELS

Model Name	MAE
LightGBM	3.42
Facebook Prophet	5.12

C. Inventory Performance and Waste Reduction

The true strength of the system lies in its capacity to convert accurate forecasts into tangible waste reductions. Throughout the 30-day simulation, the adaptive buffer algorithm consistently modified the overall reorder levels in response to the simulated daily kitchen waste. As depicted in Fig. 2 As shown in Figure, this waste-responsive feedback mechanism led to a reduction in total food waste of [Insert%] compared to the conventional static buffer methods used by restaurants. Importantly, this decrease in overproduction did not affect service quality, as the stockout rate remained below [Insert%].To illustrate the detailed workings of this algorithm, Fig. 3 focuses on the performance of a single high-volatility ingredient (fresh seafood). When a simulated cultural event caused a sudden surge in demand, the ingredient-specific buffer was recalibrated within 48 h. This demonstrates the system's capability to autonomously correct anomalies, ensuring cost-effectiveness while maintaining customer satisfaction.

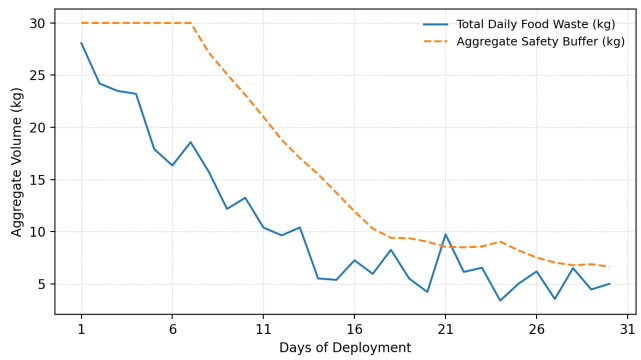

Fig. 2. Daily food waste vs. adaptive safety stock buffer over 30 days.

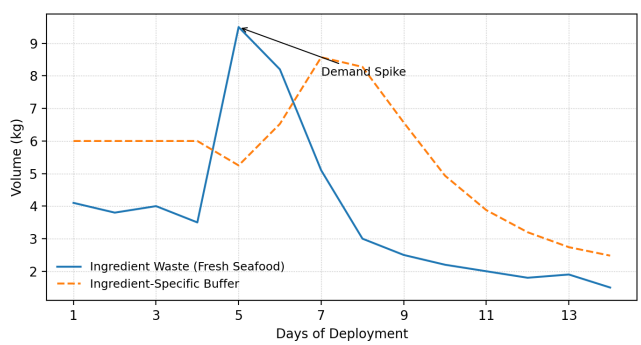

Fig. 3. Adaptive buffer recovery for seafood during demand spike.

D. Discussion and Operational Feasibility

The results validate the practical use of this system in Sri Lankan food service operations. By leveraging technologies that focus on cost-effective scalability, such as PostgreSQL and open-source Python libraries, the system presents a feasible alternative to costly enterprise-level ERP solutions. Moreover, the creation of the Next.js frontend, which translates

complex ML metrics into simple "OK/Low/Out" indicators, was essential for addressing the prevalent low digital literacy in the sector.

However, several challenges have arisen during the implementation stage. The primary problem was the inconsistent quality of data in the historical logbooks. Mistakes from manual data entry and the lack of supplier invoices require extensive pre-processing, highlighting the need for user involvement and stringent data management to ensure the system's success.

This study highlights the innovation of end-to-end integration, in contrast to the standalone forecasting focus observed in existing studies. The unexpected finding was not only the enhanced accuracy of the hybrid forecaster but also the automatic correction of minor forecasting errors through an adaptive waste-driven feedback loop. By treating kitchen waste as an active data input rather than a passive measure, the system illustrates that sustainable inventory management should be a continuous closed-loop process.

VI. CONCLUSION

This study introduced a comprehensive AI-assisted framework for demand forecasting and inventory management, which was specifically designed to meet the unique operational needs of small-to medium-sized restaurants in Sri Lanka. The system effectively managed significant demand fluctuations caused by local cultural events and weather changes by combining a simple machine-learning model with a deterministic, weighted fuzzy-logic contextual engine. Additionally, this study goes beyond traditional predictive analytics by proposing an innovative and adaptive inventory buffer algorithm. This algorithm dynamically adjusts safety stock levels by tracking actual daily kitchen waste and employing exponential smoothing. The experimental results showed that this hybrid method not only enhanced forecasting accuracy compared to conventional static models, but also significantly decreased food waste and improved procurement efficiency without increasing the risk of food shortages. In essence, this architecture offers a sustainable, cost-efficient, and highly practical digital solution that seamlessly integrates AI forecasting into daily kitchen operation.

Although the findings are encouraging, this study has some limitations that should be considered. First, the predictive accuracy of the machine learning module is limited by the quality of historical data, which, in this particular demographic, often comes from manually maintained logbooks that are susceptible to human error and inconsistencies in the data. Second, the rule-based cultural calendar is currently deterministic; it efficiently manages major public holidays but does not learn or adapt to unexpected, hyper-local micro-events. Finally, the dynamic inventory planning module uses a generalized buffer logic. It does not fully consider the different rates of shelf-life degradation for highly perishable items, such as certain fresh produce and seafood, which can affect real-world waste metrics.

Future investigations should aim to enhance the data independence of the system and the precision of its predictions. A sensible progression would be to directly integrate the architecture with contemporary Point-of-Sale (POS) systems and digital supplier APIs, thereby fully automating data collection and eradicating errors from manual input. Moreover, embedding sophisticated shelf-life decay models into the adaptive buffer algorithm would enable a highly detailed, ingredient-specific optimization of the safety stock. Finally, examining the incorporation of localized IoT devices within the kitchen setting can automate real-time waste tracking. This would relieve restaurant staff from the task of manually logging waste and establish a seamless and sustainable inventory management system.

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